APPLIED MULTIVARIATE METHODS
833:525 / 970:527
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Homework 09 (50 points)

Problem 1:

Dataset to be used: `stress.csv`
Please do not hand in computer output separately from responses to the questions below. Instead, cut and paste relevant output. Copy and paste (only relevant) R codes at the end of your homework in an appendix (annotated and in chronological order).

In a social-behavioral study, each participant's temperature was measured at baseline, and 2, 4, and 8 hours after exposure to a social stressor task at baseline. Participants were randomly assigned to two groups at baseline. Participants were also asked to complete survey measuring acute stress at baseline. The data are defined:

Variable	Code
ID	Participant ID
temp	Participant temperature
group	0 = group A (writing task), 1 = group B (speaking task)
score	Stress score at baseline
time	0,2,4,8 hours after task

a) Researchers are interested in fitting a GEE model with temperature as outcome and time, group, and their interactions as covariates. Write the mean response of the GEE model. (15 points)

The mean function of the repeated measures model for outcome, Temperature.
E(Y_ij)= β_0 +β_1 〖time〗_ij +β_2 〖group〗_i+β_3 (〖time〗_ij* 〖group〗_i)

E(Y_ij)=100.30567 – 0.03455 〖time〗_ij – 0.09745 〖group 〗_i– 0.20156 〖time〗_ij* 〖group〗_i
H0: $\beta 3 = 0$; the trajectory of temperature over time is not different with groups (interaction term)
HA: $\beta 3 \neq 0$; the trajectory of temperature over time is different between the groups
Level of Significance Alpha = 5%
Step 3: Test Statistics: Wald test statistic associated with the interaction term: 5.615e+01
Step 4 : P value P-value: 6.73e-14 Conclusion
P-value: 6.73e-14 which is less than α, 0.05 So, we reject the H0.
At the 5% level of significance, the data provides enough evidence to conclude that the trajectory of temperature over time is different between the task groups.

b) Fit the GEE model specified in (a) and try different working correlation structures (exchangeable, AR(1), and unstructured). For each correlation structure, use a hypothesis test to investigate whether the trajectory of temperature over time is different between the task groups. In your response, write down the hypotheses, test statistic, p value, and conclusion. You do not have to interpret findings in this step. Do all the working correlation assumptions lead to the same conclusions? (15 points)

Autoregressive AR (1): It assumes correlation fades gradualy after one period
E(Y_ij)=100.3565 – 0.0358 〖time〗_ij – 0.0967 〖group 〗_i– 0.1676 〖time〗_ij* 〖group〗_i
Step 1: Hypothesis:
H0: $\beta 3 = 0$; the trajectory of temperature over time is not different between the task groups
HA: $\beta 3 \neq 0$; the trajectory of temperature over time is different between the task groups
Step 2: Level of Significance
Alpha = 5%
Step 3: Test Statistics

Wald test statistic associated with the interaction term: 3.92e+01
Step 4 : P value
P-value: 3.9e-10
Step 5 : Decision and Conclusion
P-value: 3.9e-10 which is less than α, 0.05 So, we reject the H0.
At the 5% level of significance, the data provides enough evidence to conclude that the trajectory of temperature over time is different between the task groups.

Unstructured
E(Y_ij)=100.3534 – 0.0398〖time〗_ij – 0.0457〖group〗_i– 0.2009〖time〗_ij*〖group〗_i
Step 1: Hypothesis:
H0: $\beta 3 = 0$; the trajectory of temperature over time is not different between the task groups
HA: $\beta 3 \neq 0$; the trajectory of temperature over time is different between the task groups
Step 2: Level of Significance
Alpha = 5%
Step 3: Test Statistics
Wald test statistic associated with the interaction term: 5.28e+01
Step 4 : P value
P-value: 3.7e-13
Step 5 : Decision and Conclusion
P-value: 3.7e-13 which is less than α, 0.05 So, we reject the H0.
At the 5% level of significance, the data provides enough evidence to conclude that the trajectory of temperature over time is different between the task groups. Yes, it also leads to same conclusion as for the first part. We can say that the trajectory of temperature over time is different between the task groups.

c) Which model in (b) yields the best QIC value? (5 points)
Following are the Quasi-likelihood under the Independence Criterion (QIC) step a and b however, for exchangeable model it gives is the lowest number one can say It has the best model sit compares to AR and Unstructured models.

> cat("QIC for Exchangeable model:", QIC_exch, "\n")
QIC for Exchangeable model: 3438 3432 -1712 6.92 4 3438
> cat("QIC for AR(1) model:", QIC_ar1, "\n")
QIC for AR(1) model: 3457 3451 -1721 7.1 4 3457
> cat("QIC for Unstructured model:", QIC_unstr, "\n")
QIC for Unstructured model: 3442 3436 -1714 7 4 3443
d) For the model selected in (c), interpret the slope coefficient associated with the statistical interaction. (15 points)

For exchange able models:
Here are all the Betas explanation: However, interaction term of time and group is important”

(Intercept): The estimated average value of temp when time and group are zero is 100.30567. The extremely small p-value (< 2e-16) indicates strong evidence against the null hypothesis of the intercept being zero, suggesting a significant effect.
time: The coefficient of -0.03455 suggests that temp decreases by 0.03455 units for each unit increase in time, holding other variables constant. The p-value of 0.0394 indicates this effect is statistically significant at the 5% level, although the effect size is relatively small.
group: The coefficient of -0.09745 implies that being in the treatment group (assuming group is coded as 0 for control and 1 for treatment) decreases the temp by 0.09745 units, compared to the control group, holding other factors constant. The p-value of 0.6918 suggests that this decrease is not statistically significant.
time:group: The interaction term has a coefficient of -0.20156, indicating that the effect of time on temp is different for different groups. Specifically, the decrease in temp over time is more pronounced in the treatment group by an additional 0.20156 units per time unit compared to the control group. The very small p-value (6.73e-14) strongly supports the statistical significance of this interaction effect.

```
Call:
geeglm(formula = temp ~ time + group + time:group, family = gaussian(link = "identity"),
    data = stress_data, id = ID, corstr = "exchangeable")

 Coefficients:
             Estimate   Std.err      Wald Pr(>|W|)
(Intercept) 100.30567   0.16220 3.824e+05  < 2e-16 ***
time         -0.03455   0.01677 4.242e+00   0.0394 *
group        -0.09745   0.24586 1.570e-01   0.6918
time:group   -0.20156   0.02690 5.615e+01 6.73e-14 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation structure = exchangeable
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)    3.698  0.4213
  Link = identity

Estimated Correlation Parameters:
      Estimate Std.err
alpha   0.7426 0.03713
Number of clusters:   249  Maximum cluster size: 4
> # Model with AR(1) correlation structure
> Gee_Auto_reg <- geeglm(temp ~ time + group + time:group, data = stress_data, id = ID, family = gaussian(link = "identity"), corstr = "ar1")
> summary(Gee_Auto_reg)
> # Model with AR(1) correlation structure
> Gee_Auto_reg <- geeglm(temp ~ time + group + time:group, data = stress_data, id = ID, family = gaussian(link = "identity"), corstr = "ar1")
> summary(Gee_Auto_reg)

Call:
geeglm(formula = temp ~ time + group + time:group, family = gaussian(link = "identity"),
    data = stress_data, id = ID, corstr = "ar1")

 Coefficients:
            Estimate  Std.err     Wald Pr(>|W|)
(Intercept) 100.3565   0.1626 3.81e+05  < 2e-16 ***
time         -0.0358   0.0174 4.25e+00    0.039 *
group        -0.0967   0.2481 1.50e-01    0.697
time:group   -0.1676   0.0268 3.92e+01  3.9e-10 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation structure = ar1
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)     3.72   0.422
  Link = identity

Estimated Correlation Parameters:
      Estimate Std.err
alpha    0.834  0.0275
Number of clusters:   249  Maximum cluster size: 4
>
> # Model with Unstructured correlation structure
> Gee_Unstruc <- geeglm(temp ~ time + group + time:group, data = stress_data, id = ID, family = gaussian(link = "identity"), corstr = "unstructured")
> summary(Gee_Unstruc)
```

```
Call:
geeglm(formula = temp ~ time + group + time:group, family = gaussian(link = "identity"),
    data = stress_data, id = ID, corstr = "unstructured")

 Coefficients:
            Estimate  Std.err     Wald Pr(>|W|)
(Intercept) 100.3534   0.1623 3.82e+05  < 2e-16 ***
time         -0.0398   0.0172 5.34e+00    0.021 *
group        -0.0457   0.2480 3.00e-02    0.854
time:group   -0.2009   0.0276 5.28e+01  3.7e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation structure = unstructured
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)      3.7   0.421
  Link = identity

Estimated Correlation Parameters:
          Estimate Std.err
alpha.1:2    0.832  0.0470
alpha.1:3    0.730  0.0534
alpha.1:4    0.629  0.0804
alpha.2:3    0.823  0.0489
alpha.2:4    0.679  0.0672
alpha.3:4    0.722  0.0676
Number of clusters:   249  Maximum cluster size: 4
> # Assuming 'Gee_temp', 'Gee_Auto_reg', and 'Gee_Unstruc' are already fitted GEE models from your previous analyses
>
> # Assuming 'Gee_temp', 'Gee_Auto_reg', and 'Gee_Unstruc' are already fitted GEE models from your previous analyses
>
>
> # Calculate QIC for each model using the correct function
> QIC_exch <- QIC(Gee_temp)
> QIC_ar1 <- QIC(Gee_Auto_reg)
> QIC_unstr <- QIC(Gee_Unstruc)
```

```
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> # Calculate QIC for each model using the correct function
> QIC_exch <- QIC(Gee_temp)
> QIC_ar1 <- QIC(Gee_Auto_reg)
> QIC_unstr <- QIC(Gee_Unstruc)
>
> # Print QIC values to compare which model fits best
> cat("QIC for Exchangeable model:", QIC_exch, "\n")
QIC for Exchangeable model: 3438 3432 -1712 6.92 4 3438
> cat("QIC for AR(1) model:", QIC_ar1, "\n")
QIC for AR(1) model: 3457 3451 -1721 7.1 4 3457
> cat("QIC for Unstructured model:", QIC_unstr, "\n")
QIC for Unstructured model: 3442 3436 -1714 7 4 3443
>
```

```
#HW09
#Applied Multivariate
#Umer Farooq

rm(list=ls())
install.packages("lmtest")
install.packages("car")
install.packages("MASS")
install.packages("lme4")
install.packages("glmnet")
library(questionr)
library(nnet)
library(lmtest)
install.packages("geepack")
library(geepack)
setwd("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Multivariate")
stress_data <- read.csv("stress-1.csv")
# GEE Model with temperature and time
Gee_temp <- geeglm(temp ~ time + group + time:group, data = stress_data, id = ID, family = gaussian(link = "identity"), corstr =
"exchangeable")
summary(Gee_temp)

# Model with AR(1) correlation structure
Gee_Auto_reg <- geeglm(temp ~ time + group + time:group, data = stress_data, id = ID, family = gaussian(link = "identity"), corstr = "ar1")
summary(Gee_Auto_reg)

# Model with Unstructured correlation structure
Gee_Unstruc <- geeglm(temp ~ time + group + time:group, data = stress_data, id = ID, family = gaussian(link = "identity"), corstr =
"unstructured")
summary(Gee_Unstruc)
# Assuming 'Gee_temp', 'Gee_Auto_reg', and 'Gee_Unstruc' are already fitted GEE models from your previous analyses

# Assuming 'Gee_temp', 'Gee_Auto_reg', and 'Gee_Unstruc' are already fitted GEE models from your previous analyses


# Calculate QIC for each model using the correct function
QIC_exch <- QIC(Gee_temp)
QIC_ar1 <- QIC(Gee_Auto_reg)
QIC_unstr <- QIC(Gee_Unstruc)

# Print QIC values to compare which model fits best
cat("QIC for Exchangeable model:", QIC_exch, "\n")
cat("QIC for AR(1) model:", QIC_ar1, "\n")
cat("QIC for Unstructured model:", QIC_unstr, "\n")
```